

Design Rules Summary

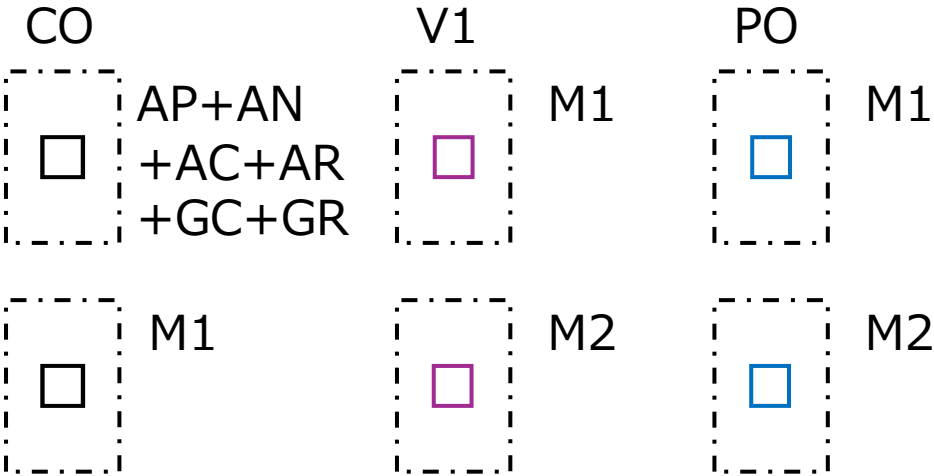
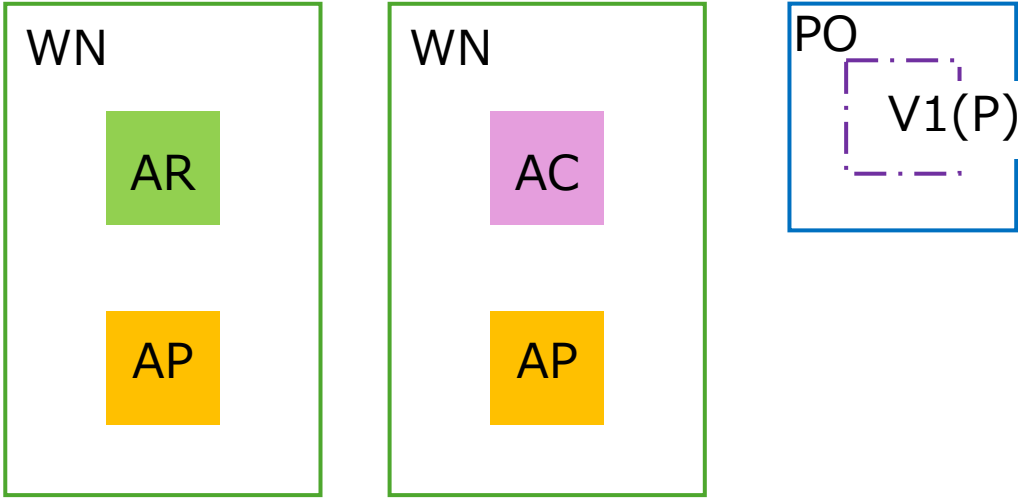
for Drawing Layers

2026/01/08

Drawing Layer Table

	Original MASK layers				Proposed Drawing layers					
	Name	NUM	TYPE		Name	NUM	TYPE			
1	PSUB	140	0		WN	140	0			
2	NW	36	0					MDP generatioed		
3	HVNW	141	0					MDP generatioed		
4	L	3	0					MDP generatioed		
					AP	3	1	Active for P+		
					AN	3	2	Active for N+		
					AR	3	3	Active for resistor		
					AC	3	4	Active for capacitance		
5	NF	25	0					MDP generatioed		
6	PF	26	0					MDP generatioed		
7	CL	143	0					MDP generatioed		
8	HPBE	144	0					No USE		
9	HNBE	145	0					No USE		
10	PBE	146	0					MDP generatioed		
11	NBE	147	0					MDP generatioed		
12	HPM	33	0					MDP generatioed		
13	RHP	148	0					No USE		
14	SG	8	0							
					GC	8	1	Gate Poly transistor		
					GR	8	2	Gate POLY resistor		
15	PM	35	0					MDP generatioed		
16	NM	7	0					MDP generatioed		
17	R	12	0					MDP generatioed		
18	PSD	9	0					MDP generatioed		
19	NSD	28	0					MDP generatioed		
20	CONT	11	0		CO	11	0			
21	M1	13	0		M1	13	0			
22	TC	19	0		V1	19	0			
23	M2	20	0		M2	20	0			
24	PRO	14	0		PO	14	0			
	TXM1	48	0		TXM1	48	0	M1 Label		
	PIN	48	1		PIN	48	1	M1 Pin		
	TXM2	49	0		TXM2	49	0	M2 Label		
	MASK	63	0		MASK	63	0	Waiver region		
	MEAS	63	1		MEAS	63	1	Measurement region		
	SCRIBE	80	0		SCRIBE	80	0	Waiver region (EDGE SEAL)		
	prBoundary	235	0		prBoundary	235	0	Cell boundary		

Prohibited

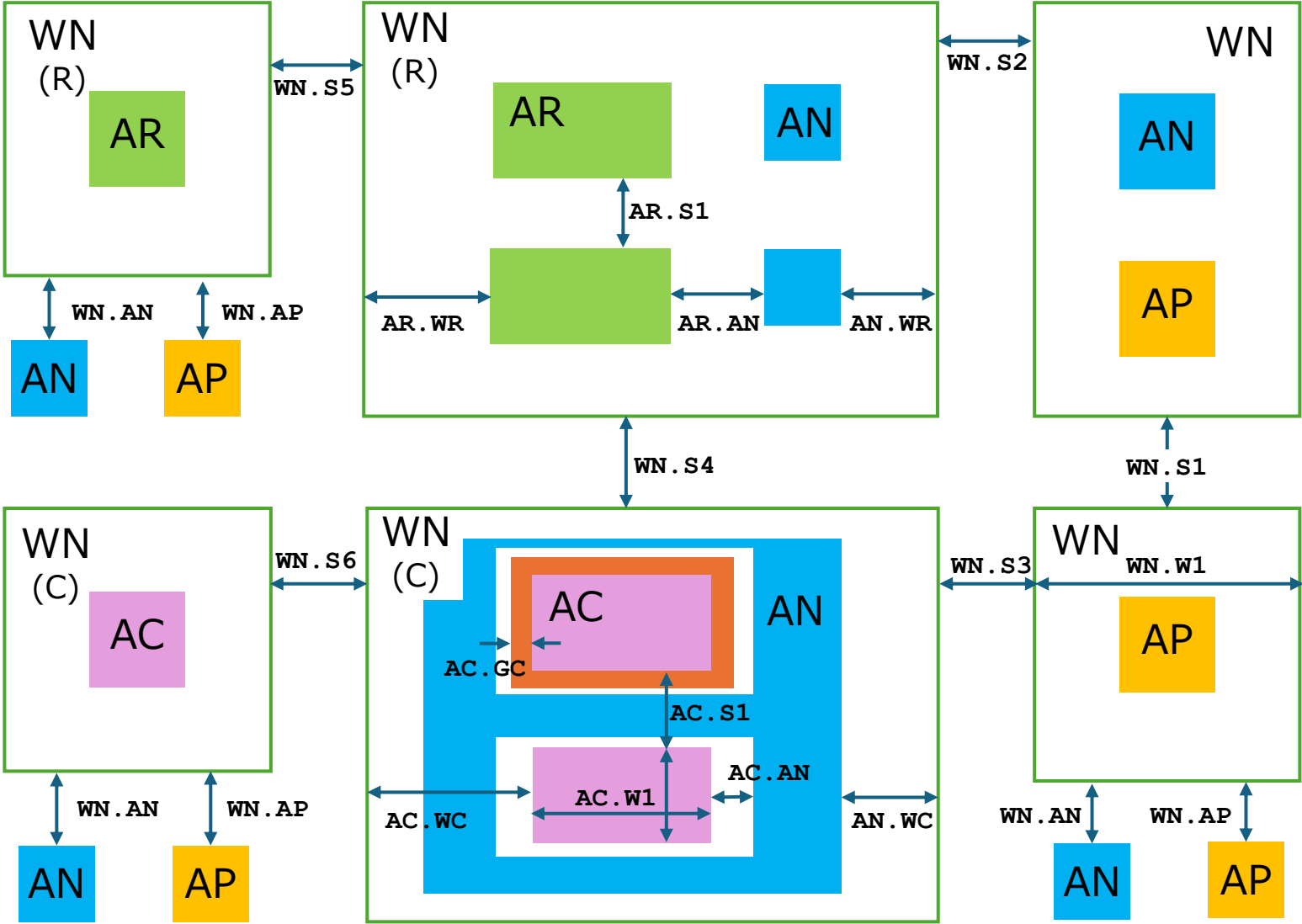


Rule	L1	L2	Func	MIN	MAX
WN.Z1	WN (R)	AP+AC	None	0.0	
WN.Z2	WN (C)	AP+AR	None	0.0	
CO.Z1	CO	AA+GC+GR	None	0.0	
CO.Z3	CO	M1	None	0.0	
V1.Z1	V1 (P)	M1	None	0.0	
V1.Z2	V1	M2	None	0.0	
PO.Z1	PO	M1	None	0.0	
PO.Z2	PO	M2	None	0.0	
PO.Z3	PO	V1	Exist	0.0	

Function:

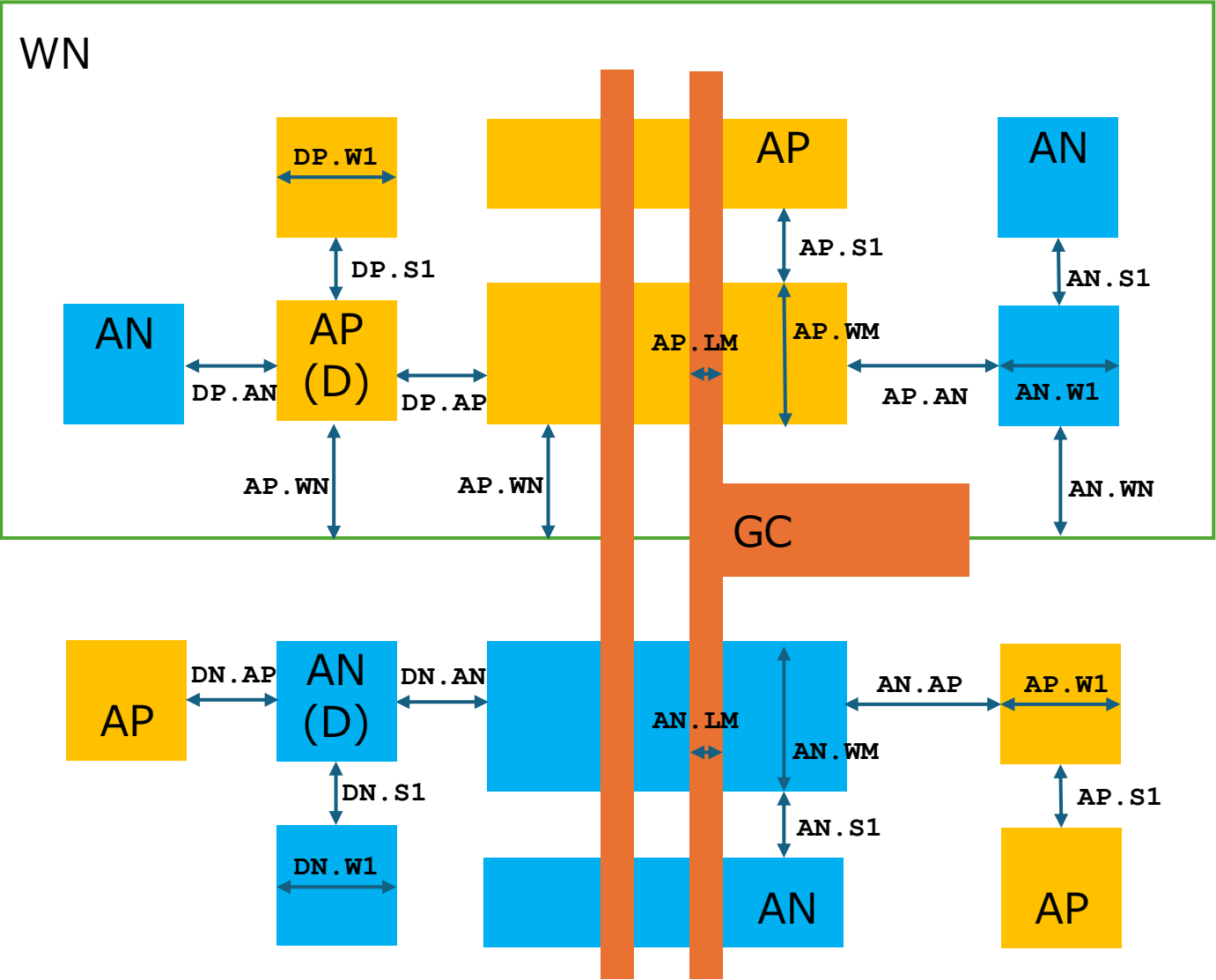
Wmin/max: L1 Width min/max
Smin/max: L1 Space min/max (L1=L2)
Smin/max: L1 Separate L2 min/max
Emin/max: L1 Enclosed L2 min/max
Fmin/max: L1 Enclosing L2 min/max
ECmin: MOS Endcap min
Wfix: L1 Windth equal
Sfix: Space equal
Donut: L1 Souuounded by L2
None: Not Exist

WN to AP/AN/AR/AC related



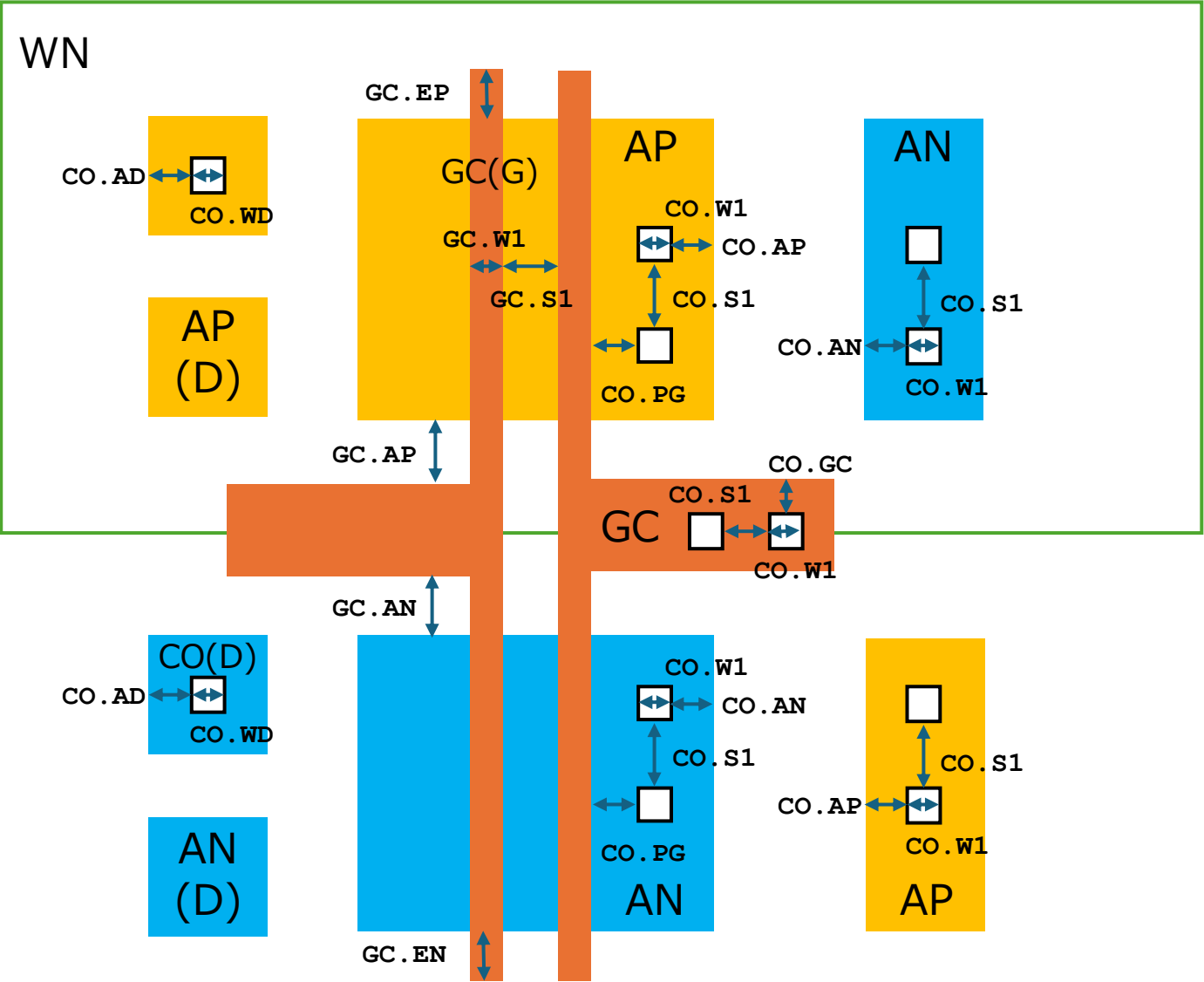
Rule	L1	L2	Func	MIN	MAX
WN.W1	WN		Wmin	8.0	
WN.S1	WN	WN	Smin	12.0	
WN.S2	WN	WN (R)	Smin	9.5	
WN.S3	WN	WN (C)	Smin	12.0	
WN.S4	WN (R)	WN (C)	Smin	9.5	
WN.S5	WN (R)	WN (R)	Smin	8.0	
WN.S6	WN (C)	WN (C)	Smin	12.0	
WN.AP	WN	AP	Smin	5.0	
WN.AN	WN	AN	Smin	10.0	
AR.S1	AR	AR	Smin	4.0	
AR.WR	AR	WN (R)	Emin	10.0	
AN.WR	AN	WN (R)	Emin	5.0	
AR.AN	AR	AN	Smin	4.0	
AC.W1	AC		Wmin/max	28.5	120.0
AC.S1	AC	AC	Smin	???	???
AC.WC	AC	WN (C)	Emin	???	???
AN.WC	AN	WN (C)	Emin	3.0	
AC.AN	AC	AN	Smin	2.8	
AC.GC	AC	GC	Emin	1.4	
AC.R1	AC	AN (C)	Donut	0.0	

MOS transistor related



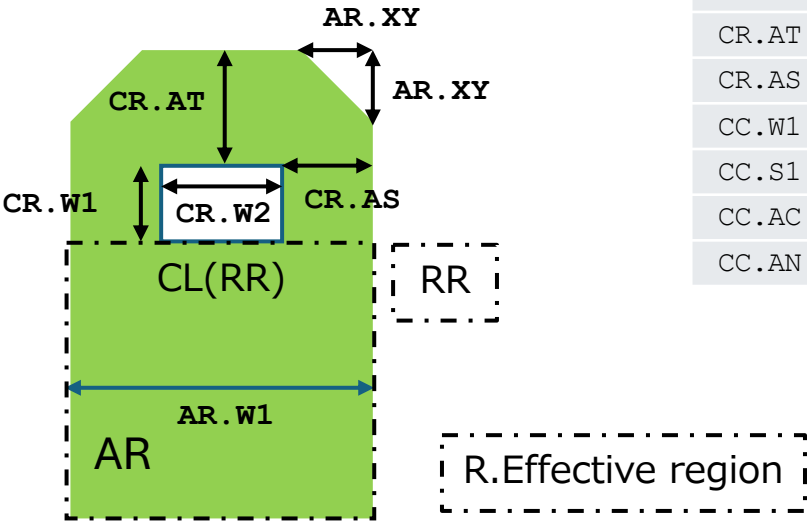
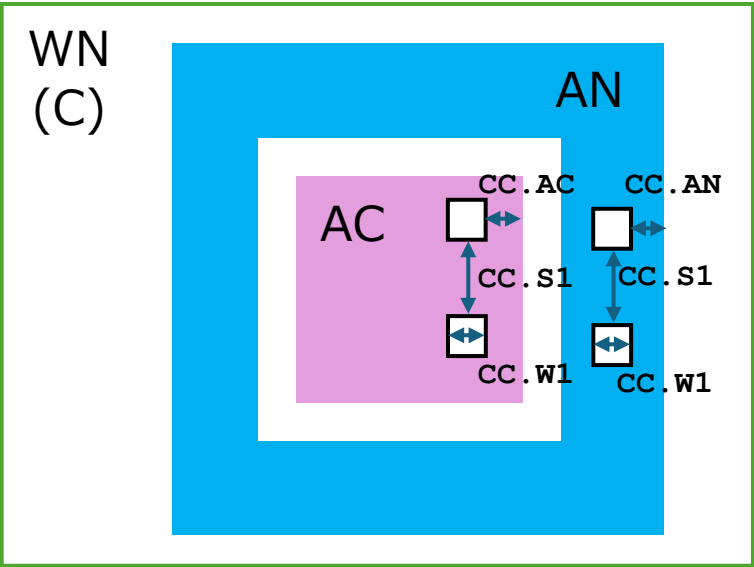
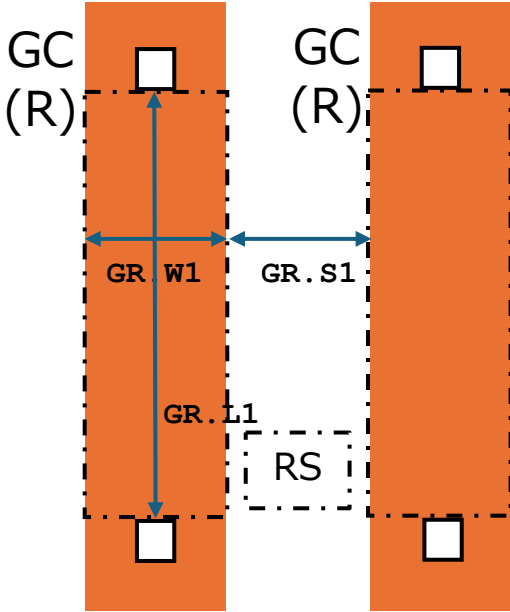
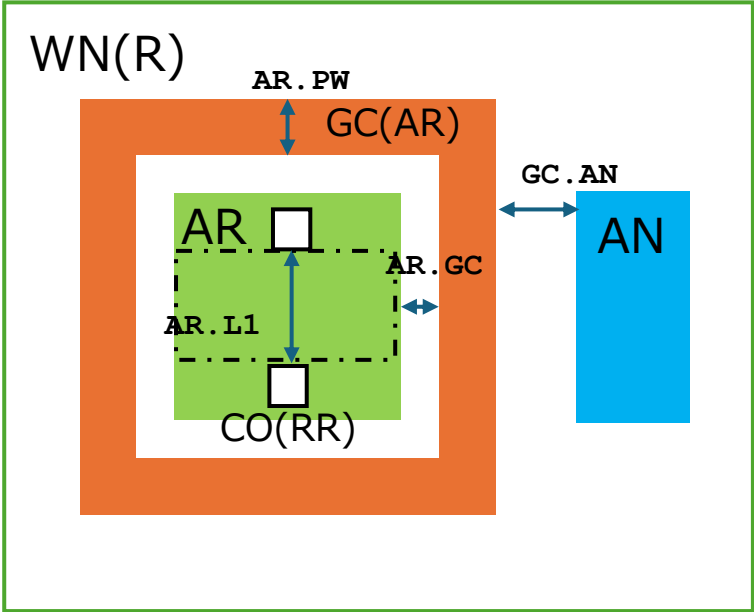
Rule	L1	L2	Func	MIN	MAX
AP.W1	AP		Wmin	1.4	
AP.S1	AP	AP	Smin	1.4	
AP.AN	AP	AN	Smin	2.8	
AP.WM	PMOS	AP-GC	Wmin/max	3.4	60.0
AP.LM	PMOS	GC-AN	Lmin/max	1.0	30.0
AP.WN	AP	WN	Emin	7.0	
DP.W1	DP		Wmin	2.8	
DP.S1	DP	DP	Smin	???	???
DP.AP	DP	AP	Smin	2.8	
DP.AN	DP	AN	Smin	2.8	
AN.W1	AN		Wmin	1.4	
AN.S1	AN	AN	Smin	1.4	
AN.AP	AN	AP	Smin	2.8	
AN.WM	NMOS	AN-GC	Wmin/max	3.4	60.0
AN.LM	NMOS	GC-AN	Lmin/max	1.0	30.0
AN.WN	AN	WN (M)	Emin	5.0	
DN.W1	DN		Wmin	2.8	
DN.S1	DN	DN	Smin	???	???
DN.AP	DN	AP	Smin	2.8	
DN.AN	DN	AN	Smin	2.8	

Contact and Poly related



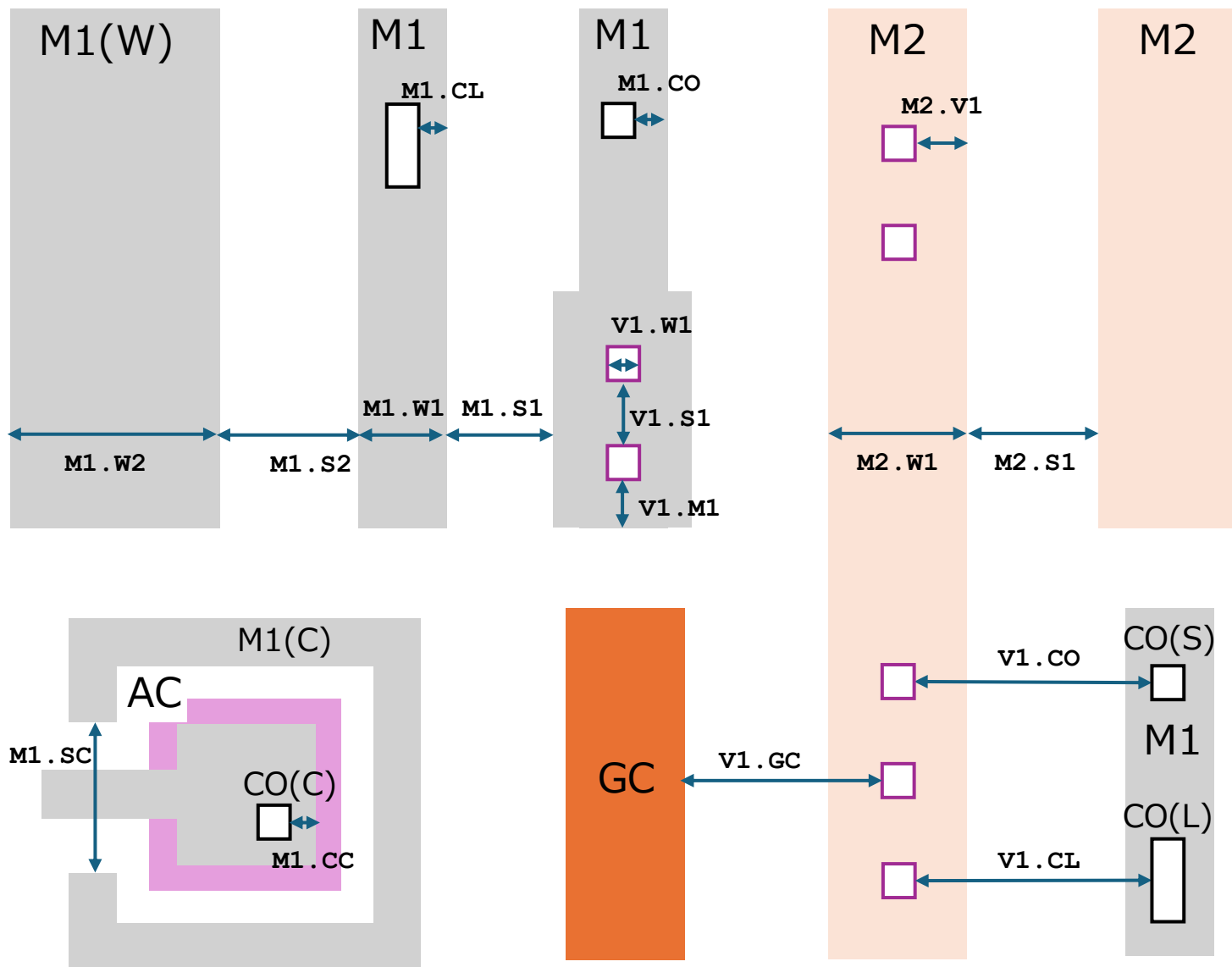
Rule	L1	L2	Func	MIN	MAX
GC.W1	GC		Wmin	1.0	30.0
GC.S1	GC	GC	Smin	1.2	
GC.AP	GC	AP	Smin	0.4	
GC.AN	GC	AN	Smin	0.4	
GC.EP	PMOS	GC	ECmin	1.2	
GC.EN	NMOS	GC	ECmin	1.2	
CO.W1	CO		Wmin	1.0	
CO.S1	CO	CO	Smin	1.0	
CO.WD	CO (D)		Wfix	1.2	
CO.AD	CO (D)	AP+AN	Emin	1.2	
CO.AP	CO	AP	Emin	0.8	
CO.AN	CO	AN	Emin	0.8	
CO.PG	CO	GC (G)	Smin	1.0	
CO.GC	CO	GC+GR	Emin	0.8	

Resistor(AR/RS) and Capacitor(AC) related



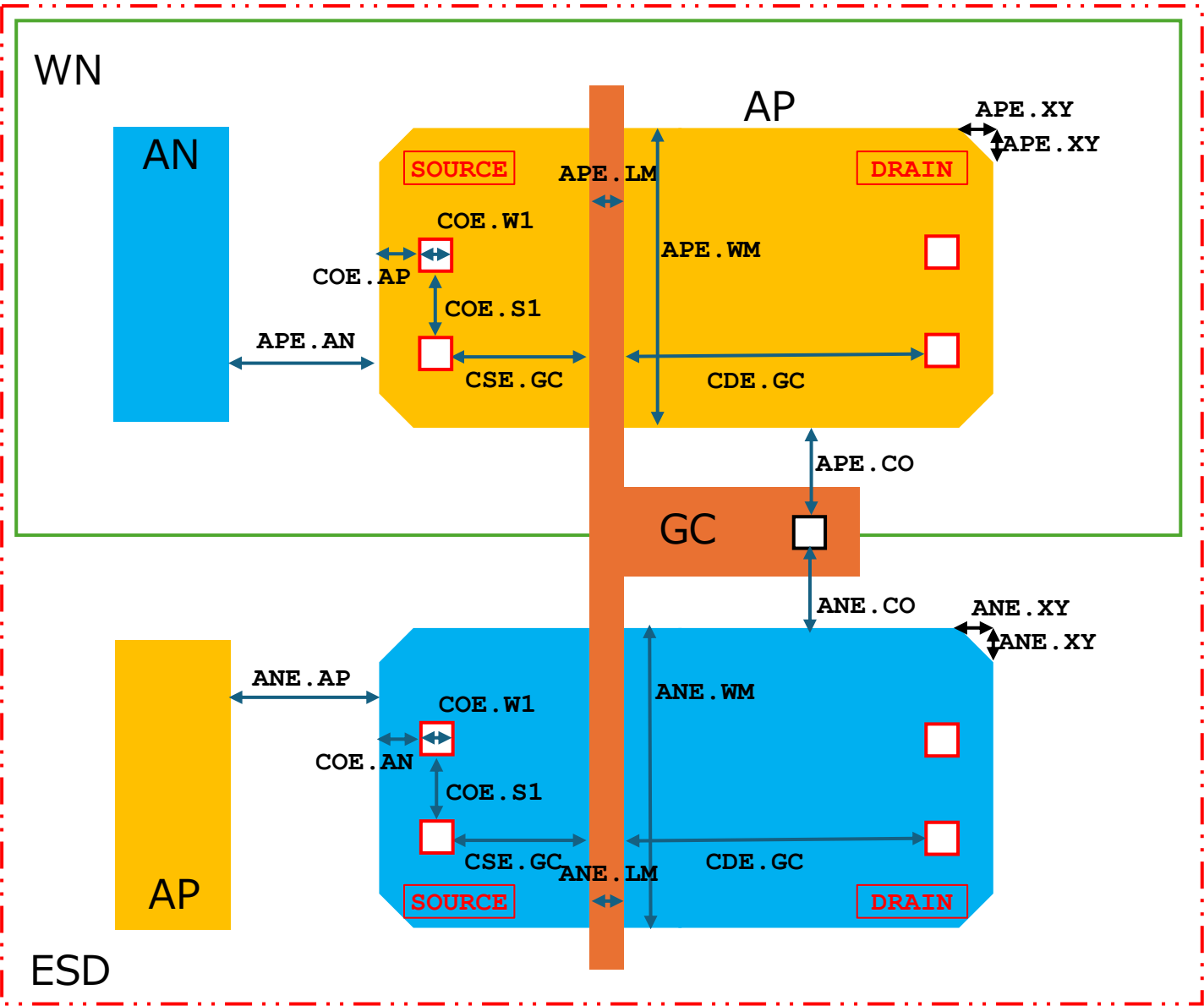
Rule	L1	L2	Func	MIN	MAX
AR.W1	RR	RR (L)	Wmin/max	2.8	20.0
AR.L1	RR	RR (W)	Lmin/max	13.0	100.0
AR.GC	GC	AR	Smin	1.0	
AR.PW	GC (AR)		Wmin	2.0	
AR.XY	AR	Bevel	XYmin	1.0	
GC.AN	GC (AR)	AN	Smin	???	???
GC.R1	AR	GC (AR)	Donut	0.0	
GC.R2	GC	TieDown		0.0	
GR.W1	RS	RS (L)	Wmin/max	4.0	20.0
GR.L1	RS	RS (W)	Lmin/max	20.0	100.0
GR.S1	RS	RS	Smin	2.0	
CR.W1	CO (RR)		Wmin	1.0	
CR.W2	CL (RR)		Wmin/max	1.0	97.5
CR.AT	CO (RR)	AR (T)	Emin	1.5	
CR.AS	CO (RR)	AR (S)	Emin	1.0	
CC.W1	CO (C)		Wmin	1.2	
CC.S1	CO (C)	CO (C)	Smin	1.2	
CC.AC	CO (C)	AC	Emin	2.9	
CC.AN	CO (C)	AN	Emin	1.2	

M1 / V1 / M2 related



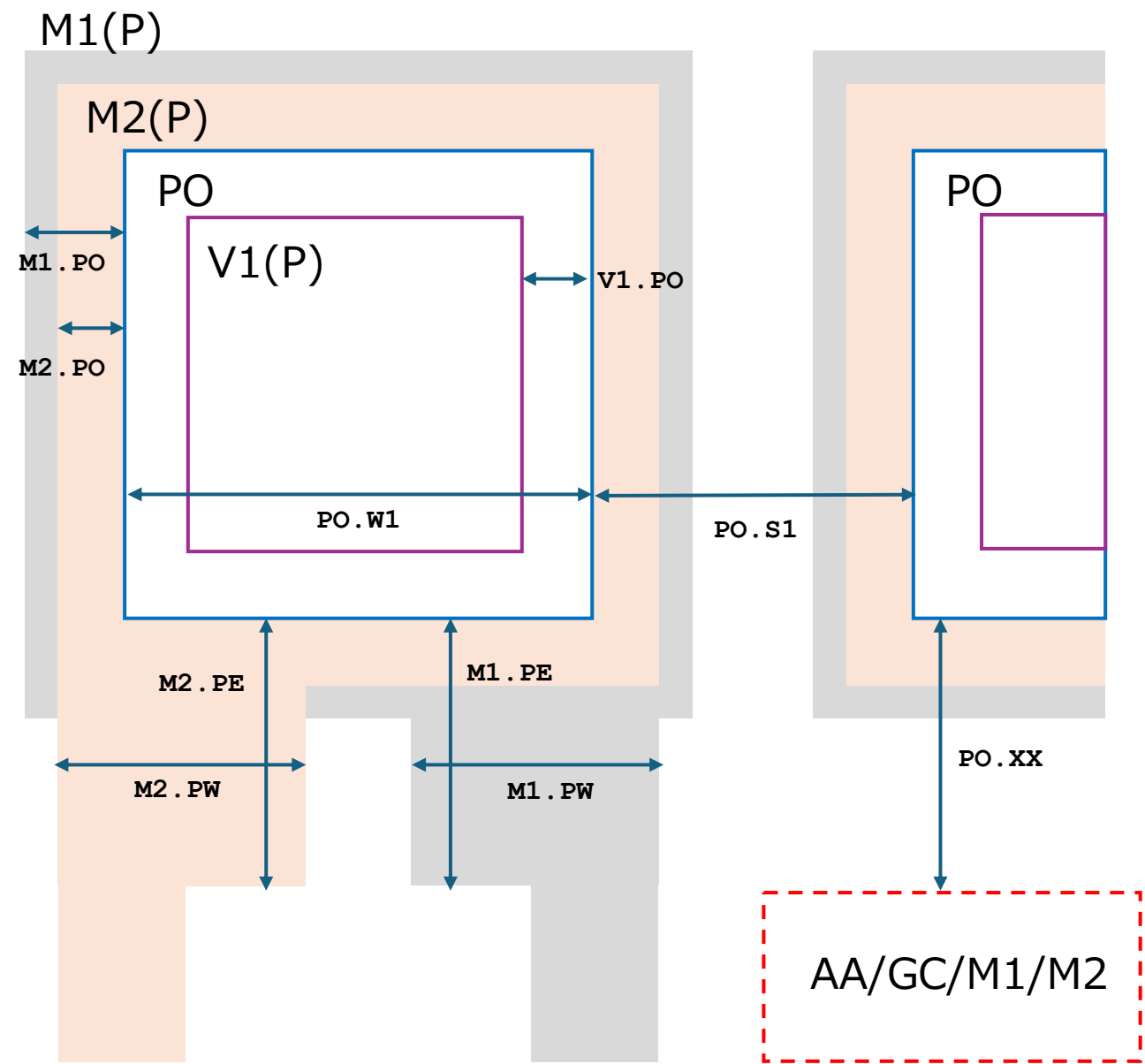
Rule	L1	L2	Func	MIN	MAX
M1.W1	M1		Wmin	1.8	
M1.S1	M1	M1	Smin	1.4	
M1.SC	M1 (C)	M1 (C)	Smin	10.0	
M1.CO	M1	CO (S)	Fmin	0.8	
M1.CC	M1	CO (C)	Fmin	1.2	
M1.CL	M1	CO (L)	Fmin	1.2	
M1.SW	M1 (W)	M1	Smin	2.0	
V1.W1	V1		Wfix	1.4	
V1.S1	V1	V1	Smin	1.5	
V1.M1	V1	M1	Emin	1.0	
V1.GC	V1	GC	Smin	1.2	
V1.CO	V1	CO	Smin	1.0	
V1.CL	V1	CO (L)	Smin	1.4	
M2.W1	M2		Wmin	3.0	
M2.S1	M2	M2	Smin	2.0	
M2.V1	M2	V1	Fmin	1.0	

ESD MOS transistor related



Rule	L1	L2	Func	MIN	MAX
APE.WM	PMOS	AP-GC	Wmin/max	11.0	60.0
APE.LM	PMOS	GC-AN	Lmin/max	2.0	30.0
APE.AN	APE	AN	Smin	10.0	
APE.XY	APE	Bevel	XYmin	1.4	
APE.CO	APE	CO	Smin	2.0	
ANE.WM	NMOS	AN-GC	Wmin/max	11.0	60.0
ANE.LM	NMOS	GC-AN	Lmin/max	2.0	30.0
ANE.AP	ANE	AP	Smin	10.0	
ANE.XY	ANE	Bevel	XYmin	1.4	
ANE.CO	ANE	CO	Smin	2.0	
COE.W1	CO (E)		Wfix	3.0	
COE.S1	CO (E)		Smin	1.6	
CDE.GC	CD (E)	GC	Smin	7.0	
CSE.GC	CS (E)	GC	Smin	3.0	
COE.AP	CO (E)	AP	Emin	4.0	
COE.AN	CO (E)	AN	Emin	4.0	

PAD related



Rule	L1	L2	Func	MIN	MAX
PO.W1	PO		Wmin	70.0	
PO.S1	PO		Smin	64.0	
M2.PO	M2 (P)	PO	Fmin	5.0	
M1.PO	M1 (P)	PO	Fmin	5.0	
PO.V1	PO	V1 (P)	Emin	10.0	
PO.AP	PO	AP	Smin	14.0	
PO.AN	PO	AN	Smin	14.0	
PO.GC	PO	GC	Smin	14.0	
PO.M1	PO	M1	Smin	14.0	
PO.M2	PO	M2	Smin	14.0	
M2.PE	M2	Extention		19.0	
M2.PW	M2	Extention	Wmin	40.0	
M1.PE	M2	Extention		19.0	
M1.PW	M2	Extention	Wmin	40.0	

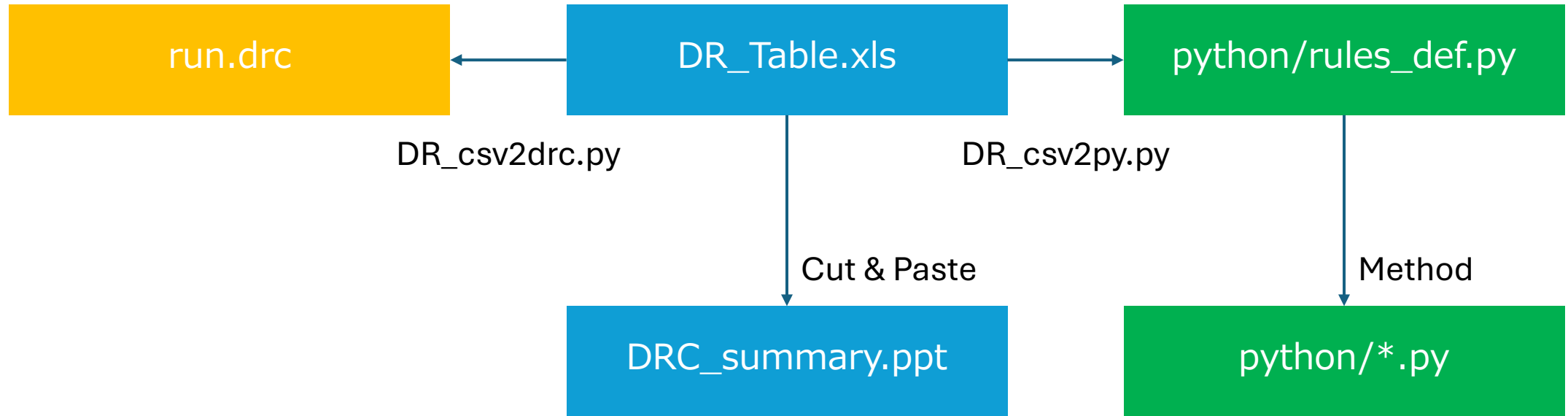
Design Rule control

DRC integrity

DRC

Document

PCell

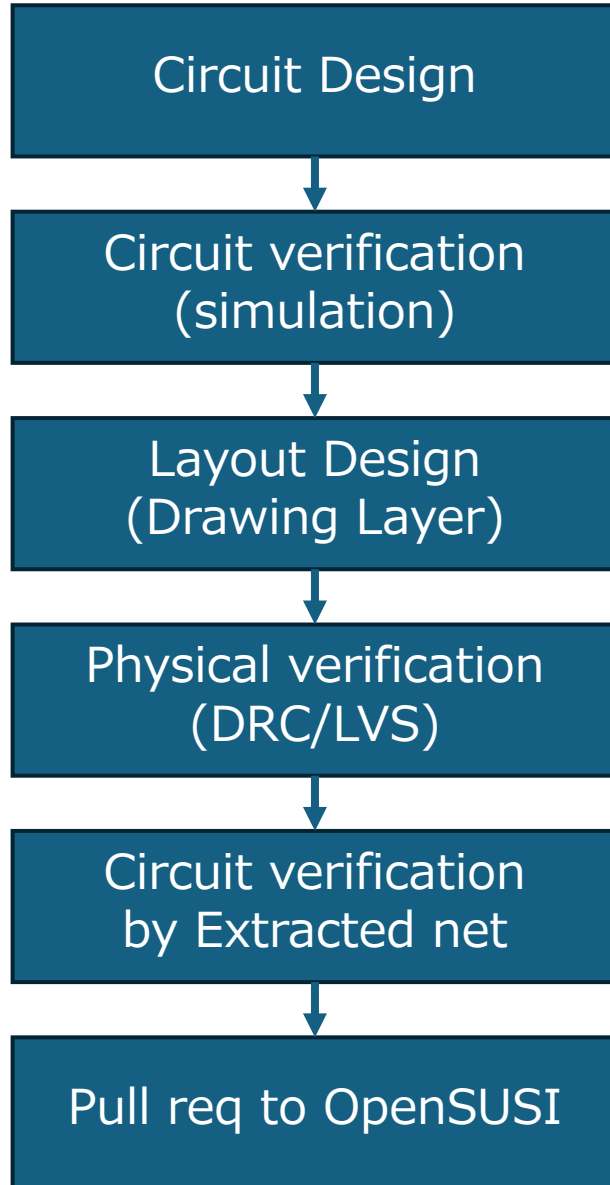


ALL Critical Dimension parameters are referenced by 'XX.YY' rule's name.

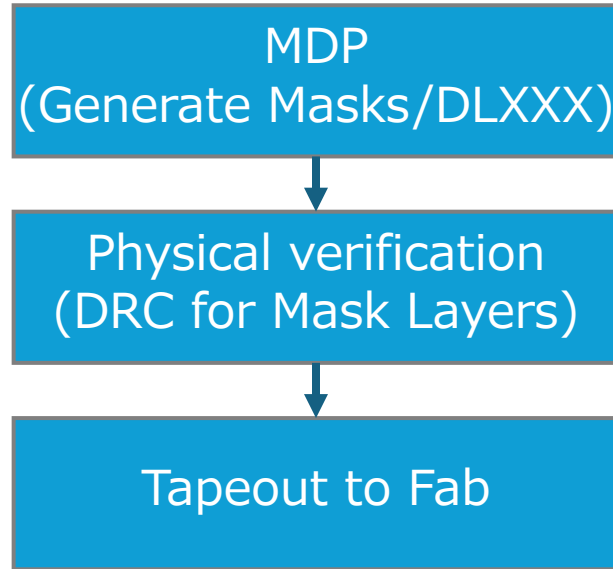
Design to Tapeout Process

Mask Data Preparation

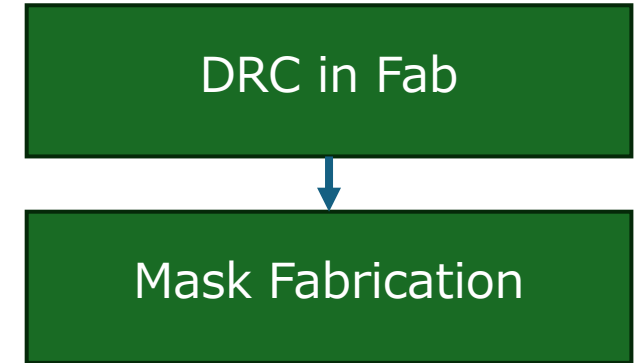
Designer



OpenSUSI



Fab



質問・確認事項

2026/01/07

DIODE

リファレンス・マニュアルTable I-1-1によればダイオードはP/NどちらもPM+NMインプラと記載

DP : PM+NM

DN : PM+NM

リファレンス・マニュアル図6.1/6.2にも同様の指示あるが、

リファレンス・マニュアル図6.12/6.13に(ESD)では

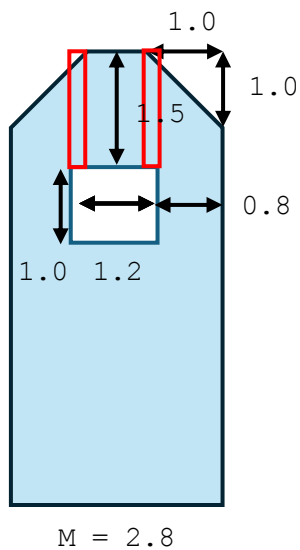
DP : NM

DN : PM

と指示がある。本当はどちらでしょう？違う場合は、Spiceモデルを公開頂けると幸いです。

また、ESD用NMOSのSPICEモデルも公開頂けると幸いです。

Design Rule for RR details



1. RR抵抗幅2.8umの場合：

ERRR14: 0.8um

RRのときは可変長コンタクト

ERRR08 : 1.5um

赤枠のエリアで ERR08 がエラー

* 角カットを1.0>0.8に変更しては？

Design Rule for V1 = TC

リファレンス・マニュアルには、

TC—CO Space ER1506/07

TC—SG Space ER1508

の記載があるが、

TC—L の記載がない、TCはLの上にあってもよいのか？

つまり、M1-V1-M2 を S/D の 上で繋いでも良いか？

CSIO

リファレンス・マニュアルには、

ERCSIO06 M1(CSIO CL) SP MUST 10.0 i 10.0

の記載があるが、

M1スペースが、MUST な理由は？

DRCでチェックすべき本質的なルールは？

6.11 CSIO
CSIOの素子内デザインルールを示す。

*注 名称
は、素子セル内の範囲を示す。

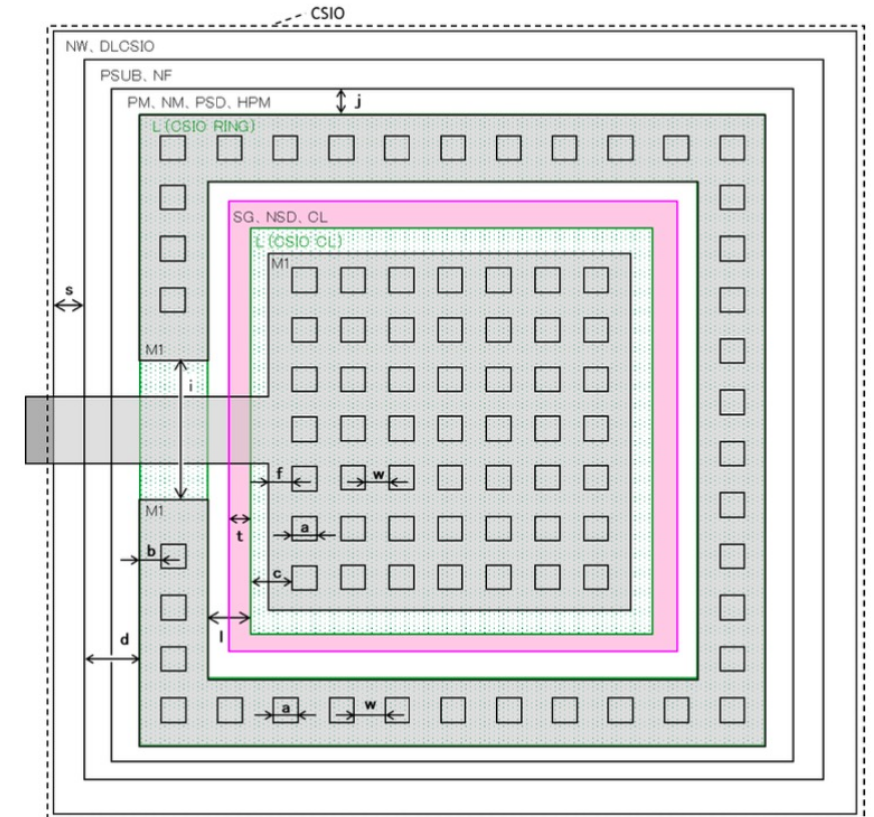


図 I-6-11. CSIOデザインルール

ESD

リファレンス・マニュアルには、

Wmin/Lmin 11.0/2.0

の記載があるが、

Wmax/max の記載がない、PMOS/NMOS と同じ 60.0/30.0 で良いか？